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Internal structure of Tenerife (Canary Islands) based on gravity, aeromagnetic and volcanological data

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Abstract

Gravity and magnetic methods have been applied to the Tenerife Island, to provide new information about its internal structure. For this study, 365 gravity stations covering the central part of the island have been selected. The anomalous density maps at different depths were obtained by means of an inversion global adjustment, on fixed density contrast, to describe the three-dimensional (3D) geometry of the anomalous bodies. On the other hand, several analysis techniques, such as reduction to the pole, spectral analysis, low-pass filtering, terrain correction and forward modelling, were applied to process the high-resolution data obtained in an aeromagnetic survey, completed with marine and terrestrial data.

The joint analysis of gravity and magnetic anomalies has shown tectonic and volcanic features that define some fundamental aspects of the structural framework and volcanic evolution of the island. A strong gravity anomaly produced by a large and deep source has been associated with an uplifted block of the basement beneath the southern part of Tenerife. The sources of the observed gravity highs from 8 km b.s.l. may be associated with the growth of the submarine shield stage that was clearly controlled by regional tectonic.

The long-wavelength magnetic anomalies reveal highly magnetic sources, interpreted as gabbro-ultramafic cumulates associated with the root zone of a large dyke swarm. This intrusive body could be topped by the emplacement zone of magma chambers that correlate with a magnetic horizon at 5.7 ± 0.8 km depth. Rooted in this highly magnetic zone, two dike–like structures can be associated with the magmatic feeding system of large recent basaltic volcanoes. A shallow magnetic horizon (1.4 km a.s.l.) can be correlated with the bottom phonolites of the Las Cañadas Edifice.

In the central part of the island the coincidence of some gravity and magnetic lows is consistent with the presence of low-density and low-magnetic materials, that infill a collapsed caldera system. The structures close to the surface are characterised by low-density areas connected with the recent volcanism, in particular the minimum over the Teide volcano. Hydrothermal alteration is assumed to be the cause of a short-wavelength magnetic low over the Teide volcano. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: aeromagnetic; gravity; volcanology; Canary Islands; Tenerife

1. Introduction

Several active volcanic areas have been successfully interpreted by means of their magnetic anomalies: Piton de la Fournaise (Lénat and Aubert, 1982),

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Mt. St. Helens (Finn and Williams, 1987), the island of Hawaii (Hildenbrand et al., 1993), Unzendake (Nakatsuka, 1994). Moreover, gravity data have been widely collected to investigate the internal structure of volcanoes. These gravity surveys range from detailed modelling of summit areas (Rymer and Brown, 1986), to broad scale modelling of subvolcanic intrusions and uplifts (Williams and Finn, 1987; Loddo et al., 1989; Rousset et al., 1989). Frequently, gravity data are interpreted together with other geophysical data, especially seismic data, as the seismic velocities and densities are directly related (Zucca et al., 1982; Villaseñor et al., 1998). Combined gravity and magnetic analysis have also shown to be especially useful in investigating the deep structure of volcanic areas (see for instance Kane et al., 1976; Rapolla et al., 1989; Okubo and Shibuya, 1993; Barberi et al., 1994).

The present work sets out to establish a structural model of Tenerife, through the integration of gravity and magnetic data with volcanological and previous geophysical data. The first gravity studies in Tenerife started with the works of MacFarlane and Ridley (1968) and later by Bosshard and MacFarlane (1970). The latter authors suggested that a subvolcanic intrusion, extending to the Moho, is the source of the island gravity maximum. Ablay and Kearey (2000) interpret the source of the same anomaly as a buried mafic volcano of substantial size, containing an old mafic complex. Two microgravimetric models have been carried out for the central part of the island (Vieira et al., 1986; Camacho et al., 1991; Camacho et al., 1996). In these models the gravity lows were related with low-density materials filling collapse calderas. In the same area the gravity model produced by Ablay and Kearey (2000) suggest the existence of a northern subcaldera domain, affected by a giant landslide. The aeromagnetic data (Socías and Mezcua, 1996) have been generically analysed in two preliminary studies (Blanco, 1997; García et al., 1997).

2. Geological setting

Although there exists some controversy about the genesis of the Canary Islands, it is well accepted that the islands have worked as independent edifices (Schmincke, 1982; Araña and Ortiz, 1991). These edifices were sited on an Upper Jurassic oceanic crust (Roest et al., 1992) containing several kilometres of sedimentary deposits (e.g. Uchupi et al., 1976). Seismic studies have revealed that the crust displays different patterns beneath the islands: under the eastern islands the seismic velocity conforms to a layer model, while under the other islands it shows a continuous variation with depth, the Moho beneath Tenerife having a depth between 13 and 15 km (Banda et al., 1981). Refraction data show that the oceanic crust near Tenerife has a mean thickness of 6.41 ± 0.42 km (Watts et al., 1997), although the deep structure of Tenerife is still inadequately known. In fact, most of the available information has been extrapolated from marine surveys (Dash and Bossard, 1969; Bosshard and MacFarlane, 1970; Banda et al., 1981; Suriñach, 1986; Watts, 1994; Watts and Masson, 1995; Ranero et al., 1995; Funk et al., 1996; Watts et al., 1997) because the scarcity of on land geophysical studies (MacFarlane and Ridley, 1968; Ortiz et al., 1986; Del Pezzo et al., 1997).

Like most oceanic islands, the emerged part represents less than 10% of the total volume of the volcanic edifice. Funk et al. (1996), using reflection seismic investigations, located the feather edge of the east flank of Tenerife at a depth of 5190 ms TWT, although the central part should be 1 or 2 km deeper because of the lithospheric flexure cause by the volcanic load (Watts, 1994). The age attributed by Funk et al. (1996) to the submarine shield base is close to that of the old subaerial massifs, indicating rapid growth of the submarine shields, and that the submarine portions of Tenerife are younger in the north–east than in the south, as are the corresponding subaerial parts. Several voluminous submarine aprons of debris avalanche deposits lie off the island's north and south flanks (Watts and Masson, 1995; Palomo et al., 1997). The subaerial part of Tenerife was originally constructed by fisural eruptions of ankaramite, basanite and alkali basalts that occurred between 12 and 3.3 Ma (Ancochea et al., 1990). The products of these eruptions are known as the Old Basaltic Series (Fúster et al., 1968). These formations made up shield volcanoes composed of a large number of superimposed lava flows crossed by numerous dikes, remains of which are cropping out at Teno (NW), Anaga (NE) and Roque del Conde (S) (Fig. 1). Probably, during the shield stage other edifices were constructed, but

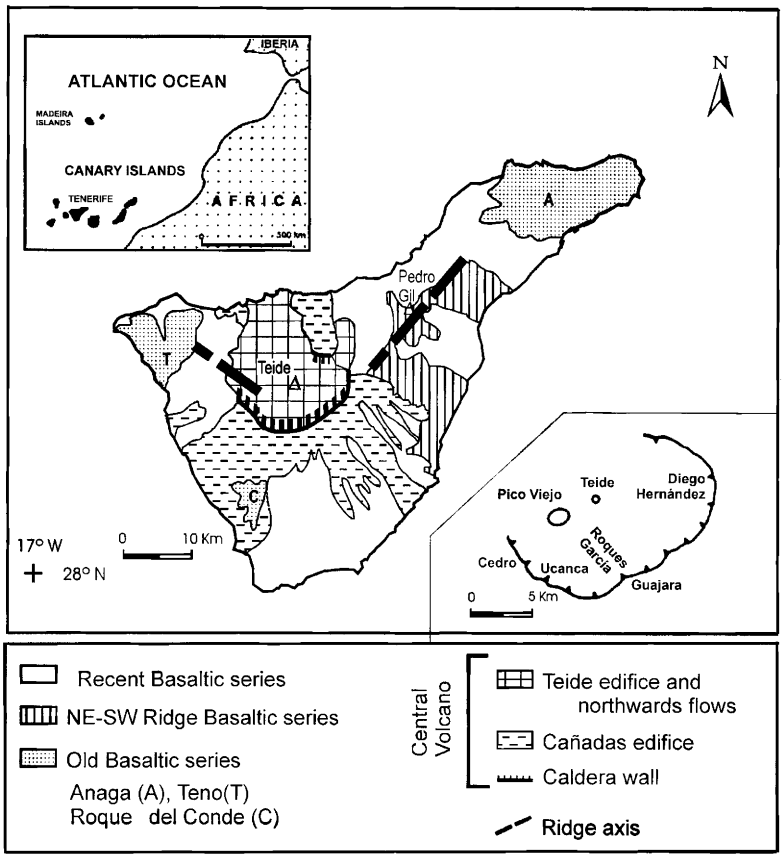

Fig. 1. Simplified geological map of Tenerife and detail of the Las Cañadas caldera. A large crest (Roques de García) separates the caldera into two zones, the western depression being older and 150 m deeper than the eastern one. The maximum height of the caldera wall is reached at Guajara (2717 m). The easternmost part, known as Diego Hernández, corresponds to the last collapse episode that occurred 179,000 years ago.

they were eroded out or are buried by the post-shield materials (see Romero et al., 2000).

In the central part of the island, from 3.5 Ma to the present, the emission of basalts and differentiated volcanics gave rise to a large central volcanic complex: The Las Cañadas edifice (Martí et al., 1994). The evolution of this edifice has been linked to the development of shallow magma chambers, evidenced by the existence of evolved phonolitic magmas. The Las Cañadas volcano culminates in a large elliptical depression measuring $16 \times 9\ km^3$, known as the Las Cañadas caldera (Fig. 1). The caldera wall is visible for 27 km along the SW, S–SE and NE sectors. During the evolution of this central volcanic complex, several periods of phonolitic volcanic activity separated by periods of mafic volcanism have been identified (Martí et al., 1994, 1995).

In the northern sector of Las Cañadas caldera, the Teide–Pico Viejo complex was constructed as the product of the most recent phase of central volcanism on Tenerife. Teide and Pico Viejo are two large stratovolcanoes that have grown in close proximity during the last 175 Ky. They overlap to form an elongated double edifice, their summits being separated by a distance of 2.5 km. The highest altitude corresponds to the youngest summit of Teide, at 3718 m. The main cone is approximately circular with a basal diameter of about 5 km. The western part of the Las Cañadas caldera is occupied by Pico Viejo products, mainly mafic to intermediate lava flows. Basaltic and phonolitic products of Teide filled the central caldera and

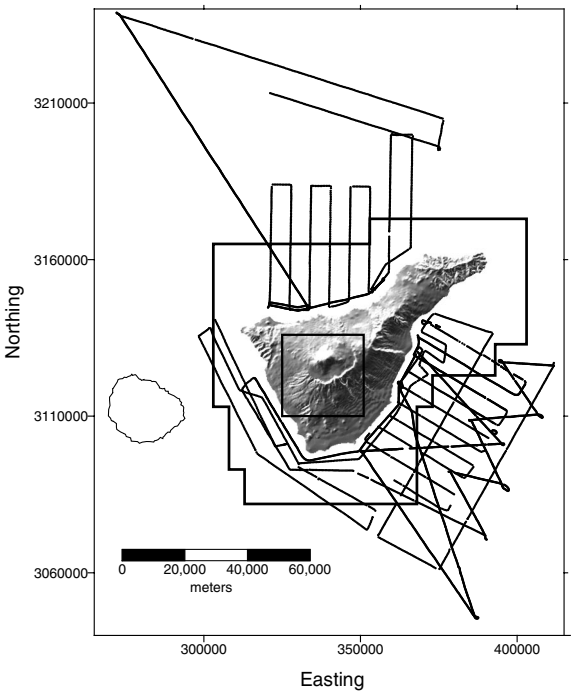

Fig. 2. Location of the magnetic and gravity surveys superimposed on the shaded-relief of Tenerife. The outer polygon delimits the area covered by the aeromagnetic survey, while the inner square marks the central zone where the gravity stations are closer and the magnetic data acquisition was carried out with a greater spatial resolution. The lines correspond to the marine magnetic profiles also considered in this study. The island that appears to the west of Tenerife is La Gomera. Coordinates are in meters referred to the Huse NUTM 28.

flowed northward to bury the slopes of the Las Cañadas edifice (Fig. 1). The last phonolitic event in the S Teide flank was the 2000 years old subplinean eruption of Montaña Blanca (Ablay et al., 1995). The magma chamber of this eruption is supposed to be located 1 km b.s.l. although other magma chambers belonging to the Teide–Pico Viejo–Las Cañadas system seem to be located deeper in the crust (Araña et al., 1989; Ablay and Martí, 1995).

The post-shield basaltic activity which overlaps the Cañadas Edifice is mainly found on the crest and flanks of the two ridges trending NNW–SSE and NE–SW, which converge on the central part of the island (Fig. 1). The Pedro Gil volcano, with a large crater and numerous dikes, in the NE–SW axis, is the most important volcano of the Ridge Basaltic Series that show a peak of activity around 0.8 Ma. (Ancochea et al., 1990). The Recent Basaltic Series that includes several historical eruptions appear in the crest and slopes of the two ridges, but it is also widespread on the whole island (Fig. 1).

3. Data acquisition

The gravity and aeromagnetic data used in this study cover the whole island, although the data resolution is much greater in the central region, which will be studied in greater detail. Magnetic data from marine surveys have been integrated, in order to know the off-shore continuity of magnetic anomalies and also to better define some on-shore anomalies near the coast-line, mitigating the edge effects. The auxiliary character of these marine data rules out the interpretation of the offshore anomalies in this paper.

3.1. Gravity data

The initial data for this study are composed by 969 gravity stations observed over the whole island of Tenerife using a La Coste–Romberg G665 gravimeter. The distribution of the stations was conditioned by the topographic roughness. Altitude data were obtained by means of geometric levelling (about a 60%) or by means of micro-barometric levelling. For the suitable application of gravity inversion in the central area (inner square in Fig. 2), we selected 365 stations located at an interval >300 m, providing a more homogeneous representation of the anomaly field.

To calculate topographic correction for the gravity observations we have considered a digital terrain model composed by about 50,000 points. From these, 41,000 correspond to terrestrial topography (Spanish Geographic Institute), with a regular grid interval of 200 m, and the rest correspond to the non-gridded bathymetry information, up to 30 km from the island (ETOPO5U: NGDC, 1988).

3.2. Magnetic data

Magnetic data were collected in an aeromagnetic survey of the Canary Islands region carried out in October 1993 (Geographic Spanish Institute internal data). Fig. 2 shows the area covered by the data, which includes the island of Tenerife and its marine

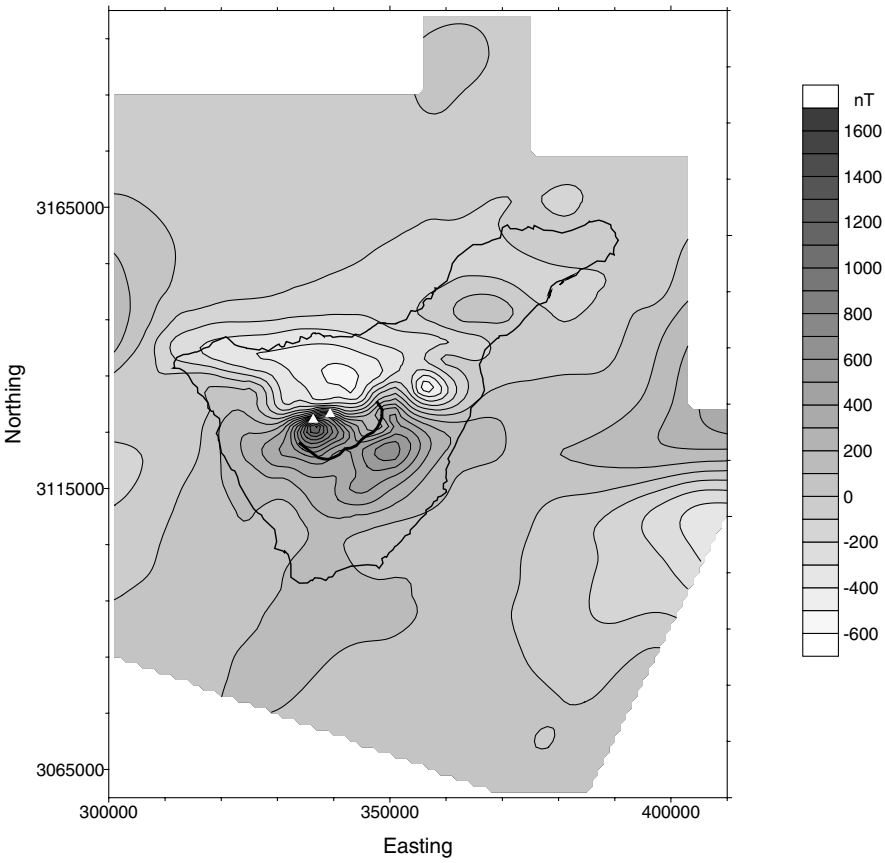

Fig. 3. Magnetic anomaly map of Tenerife obtained from aerial and marine magnetic data. The zero-anomaly isoline is slightly thicker than the rest of the isolines to make easy the distinction between positive and negative values (see the scale). As in all the magnetic maps, the black line marks the southern boundary of Las Cañadas caldera, and the white triangles show the location of the Pico Viejo (left) and Teide (right) craters.

surroundings. The flight over the external area was made at constant altitude of 3200 m. The main flight lines were flown in N–S direction with a separation between lines of 2500 m. Tie-lines were flown following the E–W direction and separated 10,000 m. The altitude of the aircraft over the inner area was 3810 m (about 100 m above the Teide peak). Lines were also flown following the N–S direction but, in this case, distance between lines was 500 m. Tie-lines were separated 5000 m and flown E–W. In both areas the spatial data resolution along each flight line was 20 m. The final grid available for this work was a 1000 m-resolution magnetic anomaly map defined at an altitude of 3810 m (Socías and Mezcua, 1996). The aeromagnetic map was combined with a marine magnetic survey carried out by the oceanographic ship Hespérides in April 1995 (Oceanographic Spanish Institute internal data). Fig. 2 shows the profiles along which the off-shore magnetic field was measured. Both marine and aeromagnetic maps show a very good agreement in the overlap regions and the final magnetic anomaly map is presented in Fig. 3.

The study has also included several terrestrial campaigns especially devoted to characterise the magnetic properties of the rocks that confirmed the previous data from paleomagnetic surveys. The magnetisation of the materials in Tenerife ranges in intensity from less than 1 A m^{-1} for the phonolitic pumices to more than 15 A m^{-1} for some massive basalts (Carracedo and Talavera, 1971; Carracedo, 1979). The highest values of NRM intensity

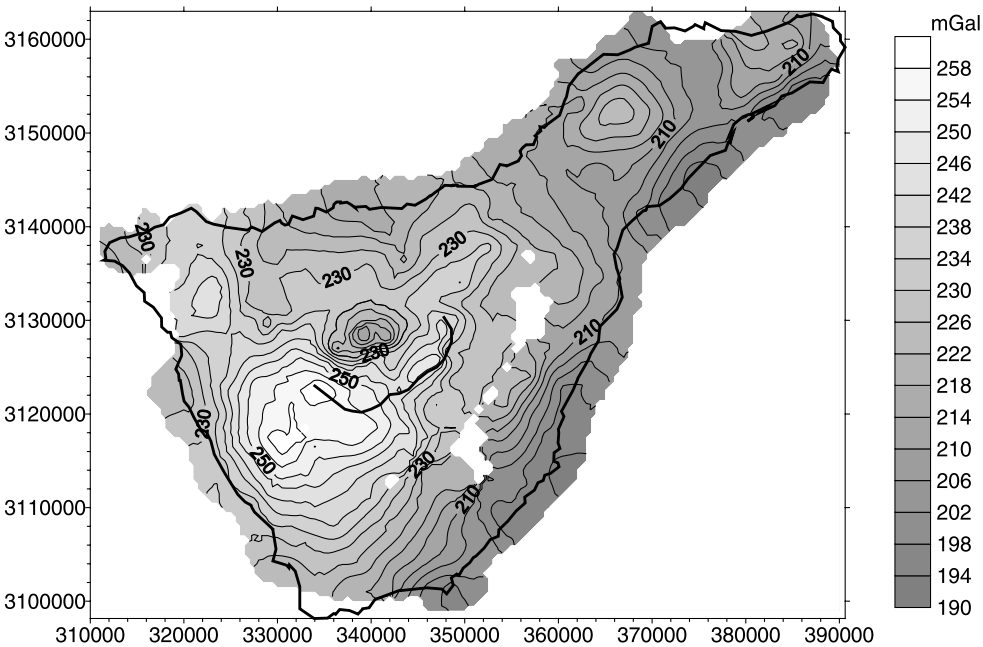

Fig. 4. Map of Bouguer gravity anomalies. The thick lines point out the coast of the island and the rim of the caldera. The white areas correspond to gaps of data. Contour interval at 4 mGal.

correspond to the recent aphanitic basalts, while ankaramitic and old basalts show lower magnetisation. On the other hand, salic rocks (phonolites and trachytes) that form lava flows, ignimbrites and pumice fall deposits, reveal the lowest NRM intensities. The average for Tenerife normally magnetised basalts is considerably higher than the average for terrestrial basalts (Prévot and Grommé, 1975), but only a little higher than the values for surface Hawaiian basalts (Hildenbrand et al., 1993).

4. Data processing

In addition to the geometric characteristics of the available gravity and magnetic data (the size of the covered area and the distance between grid nodes), the analysis and later interpretation of the gravity and magnetic anomalies were also conditioned by: (1) the volcanic nature and the age of the island (12 Ma for the oldest subaerial materials), which make the magnetisation and compaction of rocks very variable; (2) the lithology being predominantly basaltic and massive in the submarine and subaerial shield stages, and mainly felsic, with abundant pumice deposits, in the post-shield central edifice; and (3) the rugged topography of the island (altitude ranges from sea level to almost 4000 m in less than 20 km). The effect of the topography is very important and its precise quantification is essential for an appropriate interpretation.

4.1. Bouguer anomaly map and covariance analysis

Once the observations were reduced (according to the calibration values, tidal corrections, drift adjustment and offset to IGSN71 system) the corresponding Bouguer anomaly was determined. For the same, a value of 2400 kg m^{-3} was used as the mean value for the density of the topographic masses. This value is close to the values 2300, 2270 and 2480 kg m^{-3} suggested by Bosshard and MacFarlane (1970) and Watts et al. (1997) for the shallow masses, and was obtained after testing several densities and observing the resulting anomalies.

The calculated terrain correction (evaluated for distances between 3 m and 45 km by means of an on purpose process) ranges from 7 mGal (near the

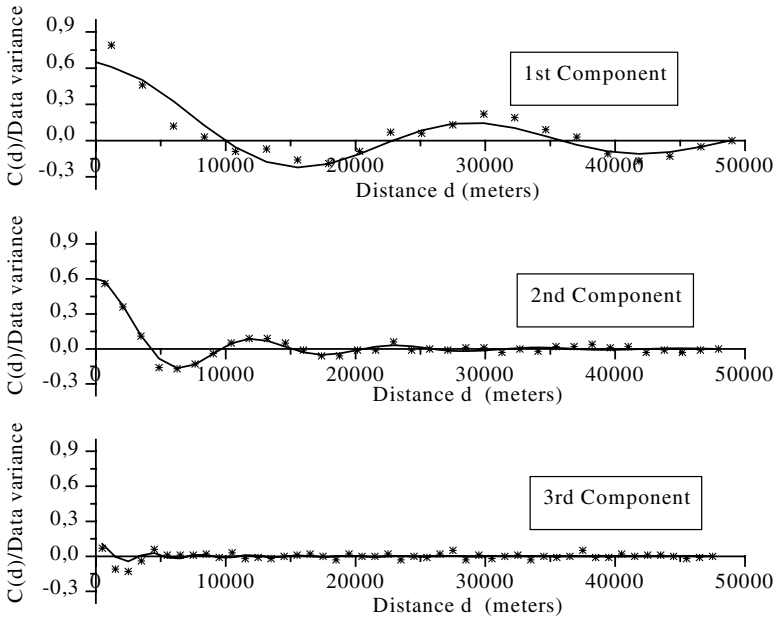

Fig. 5. Covariance functions adjusted (line) to the three successive empirical covariance distributions (stars). Analytical covariance function as $C(d) = aJ_0(cd)\exp(-bd)$ (J_0 being the zero-order Bessel function and a, b and c are empirical parameters) were used. The first two signals show decreasing amplitude and decreasing null covariance distance. The third signal is poorly significative.

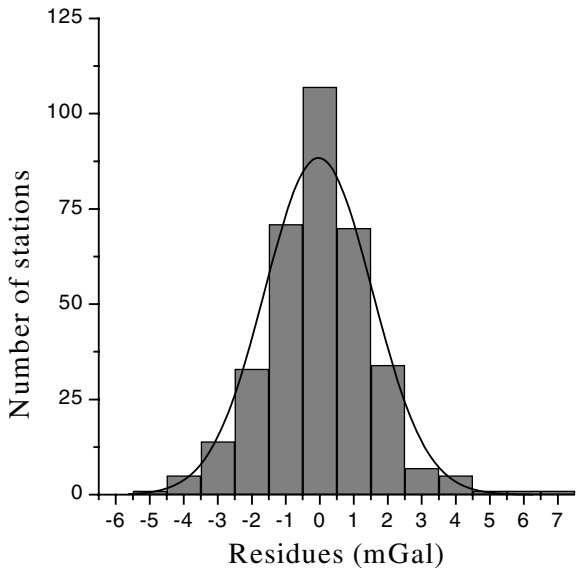

Fig. 6. Distribution of the residues resulting from successive covariance analysis. It corresponds to the noise level from the gravity data. A standard deviation of 1.58 mGal corresponds to an adjusted normal distribution (line in the figure).

coast) to 91 mGal (for the top, at 3717 m height of the Teide cone). The resulting Bouguer anomaly (Fig. 4) ranges from 190 mGal (E coast) to 260 mGal (SW of Las Cañadas). The obtained Bouguer anomaly map is very similar to the one produced by MacFarlane and Ridley (1968).

Taking into account the presence of barometric levelling and strong terrain effects, some inaccuracy can be suspected in the data Δg of the Bouguer anomaly map. For this reason, to detect observational noises and to ascertain values of the global precision of the gravity anomaly, we apply a covariance analysis and a least squares prediction, as is described by Camacho et al. (1997). The covariance analysis was applied successively and three signal levels of decreasing null covariance distance were determined. Fig. 5 shows the empirical covariances and the adjusted analytical functions, and Fig. 6 the histogram of the resulting noise distribution. According to this distribution, a standard deviation value of 1.6 mGal can be determined and adopted as a significant precision level of the gravity anomaly.

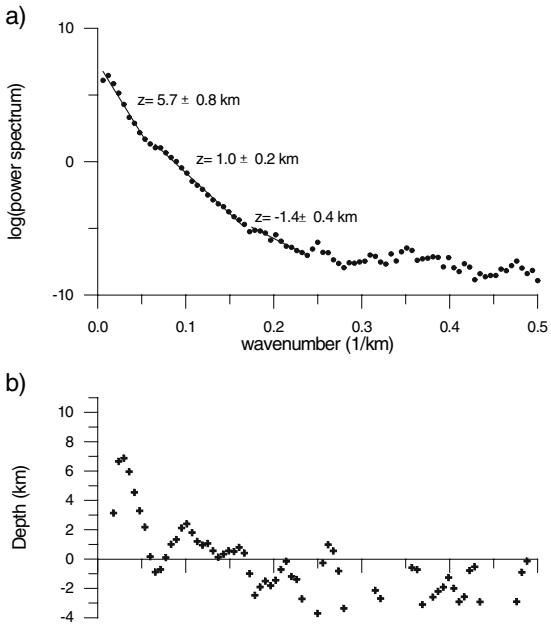

Fig. 7. Spectral analysis of the magnetic anomaly map: (a) Radially averaged power spectrum and estimation of the depth to the top of the main magnetic layers. (b) Depth to the sources as a function of wave number obtained by linear fitting of a five-point window which moves across the whole spectrum.

4.2. Magnetic data reduction

4.2.1. Spectral analysis

The spectral analysis of a magnetic anomaly map is a very useful tool for regional studies because it provides information about: (a) the amplitude and wavelength of the anomalies, making it possible to characterise quantitatively the anomalies created by regional and local features; and (b) the depth to the main magnetic layers, since the slope of the logarithm of the radially averaged power spectrum of the magnetic anomaly map as a function of wave number is directly related to the vertical distance to the main magnetic horizons (Spector and Grant, 1970). To perform this spectral analysis, the radially averaged power spectrum of the magnetic anomaly map of Tenerife was calculated and then plotted as a function of wave number (see Fig. 7a). The spectrum contains the anomalies characterised by wave numbers between 8.62×10^{-3} and $0.5\ \mathrm{km}^{-1}$, determined by the horizontal dimensions and the spatial resolution of the magnetic map. The spectrum shows three different and well-defined slopes that correspond with three magnetic horizons. The depth to each horizon was calculated by linear fitting of each section of the spectrum and is also indicated in Fig. 7a. The depth to the magnetic sources obtained through the linear fitting of a five-point window that moves across all the wave numbers is shown in Fig. 7b.

4.2.2. Reduction to the pole

To estimate the direction of the total magnetisation vector, in order to obtain a reliable reduction to the pole, the so-called "distortion analysis of the magnetic anomalies" was chosen (Fedi et al., 1994) for its straightforward excellent results (Blanco, 1997). The anomaly selected to estimate the direction of the total magnetisation vector is the most intense of those displayed in Fig. 3, that is, the one that consists of a high located to the south of the Teide–Pico Viejo complex and a low appearing to the north. Once the average direction of the total magnetisation vector was determined, the magnetic anomaly map was transformed to the corresponding reduced-to-the-pole map. From the IGRF (IAGA, 1996), it can be ascertained that the main present magnetic field ranges from -8.0 to $-8.7°$ in declination and from 37.1 to 40.8° in inclination in the area covered by the magnetic map. The values used were those corresponding to the centre of the island, that is, $D = -8°$ and $I = 38°$ that coincide with those obtained in a preliminary analysis of the central area of Tenerife (García et al., 1997).

Taking into account that large-scale deep magnetic structures are the source of long-wavelength magnetic anomalies, wavelength filtering was used to separate the effect of regional magnetic features from the effect of shallow bodies. Therefore, a Butterworth low-pass filter (Proakis and Manolakis, 1997) was applied to the magnetic anomaly map to remove the anomalies caused by shallow structures. Two different cut-off wave numbers, selected from the spectrum of Fig. 7b, were used. Firstly, the magnetic anomaly map was filtered using as the cut-off wave number $K_1 = 0.05\ \mathrm{km}^{-1}$ ($\lambda = 20$ km), considering that this is the highest wave number that fits the straight line that defines the top of the deepest magnetic layer. Thus, the map resulting from this filtering should contain only the deepest sources whose effects appear on the original magnetic map, that is, the longest wavelength

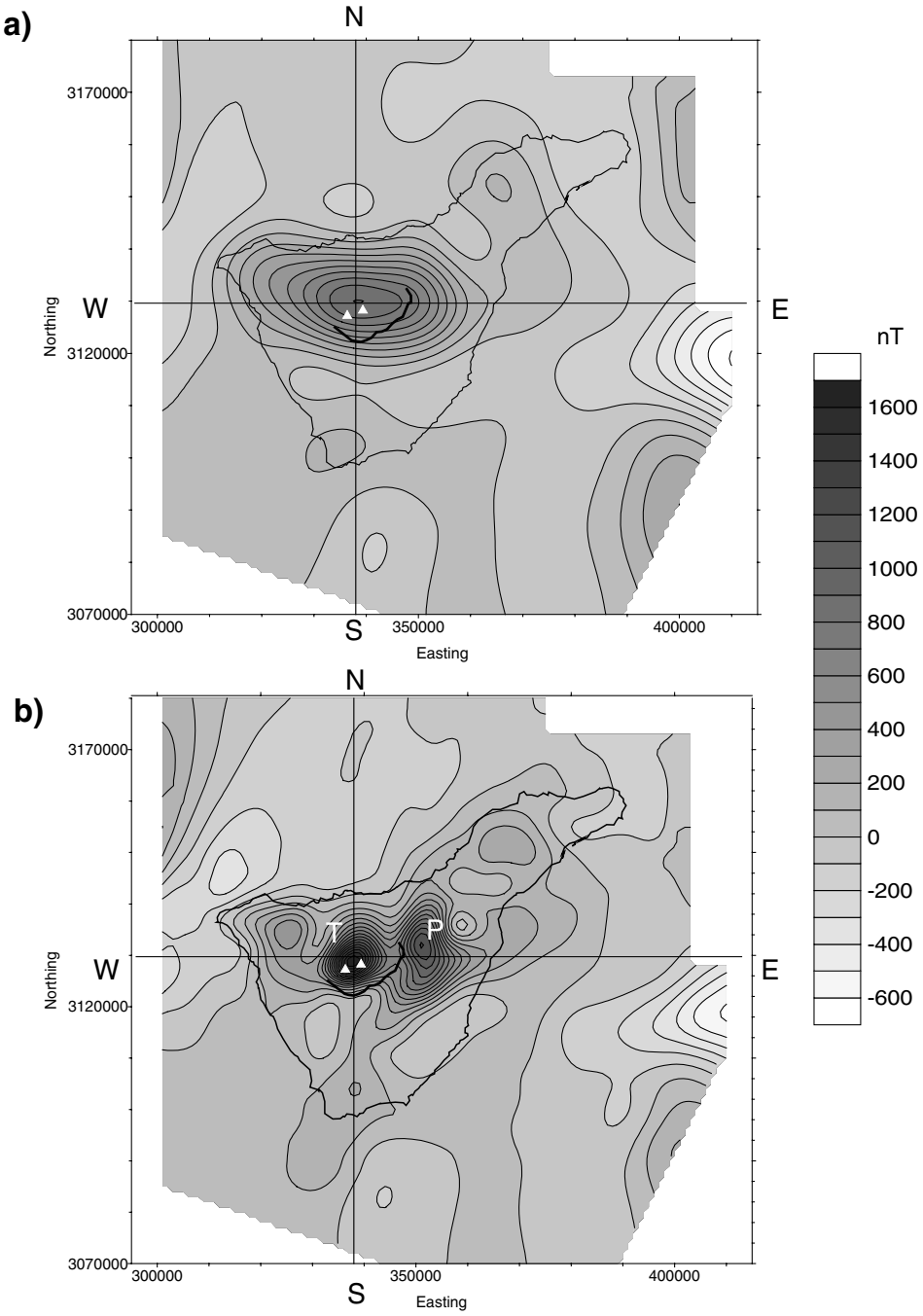

Fig. 8. (a) Reduced-to-the pole long-wavelength or regional magnetic anomaly map (obtained by low-pass filtering the original map with a cut-off wave number $K_1 = 0.05\ \mathrm{km}^{-1}$). (b) Reduced-to-the pole magnetic anomaly map where the effect of above sea level sources has been removed (obtained by low-pass filtering the original map with a cut-off wave number $k_2 = 0.17\ \mathrm{km}^{-1}$. The black lines (N–S and W–E) correspond to the profiles selected for the forward modelling of the relevant anomalies (Figs. 12 and 13).

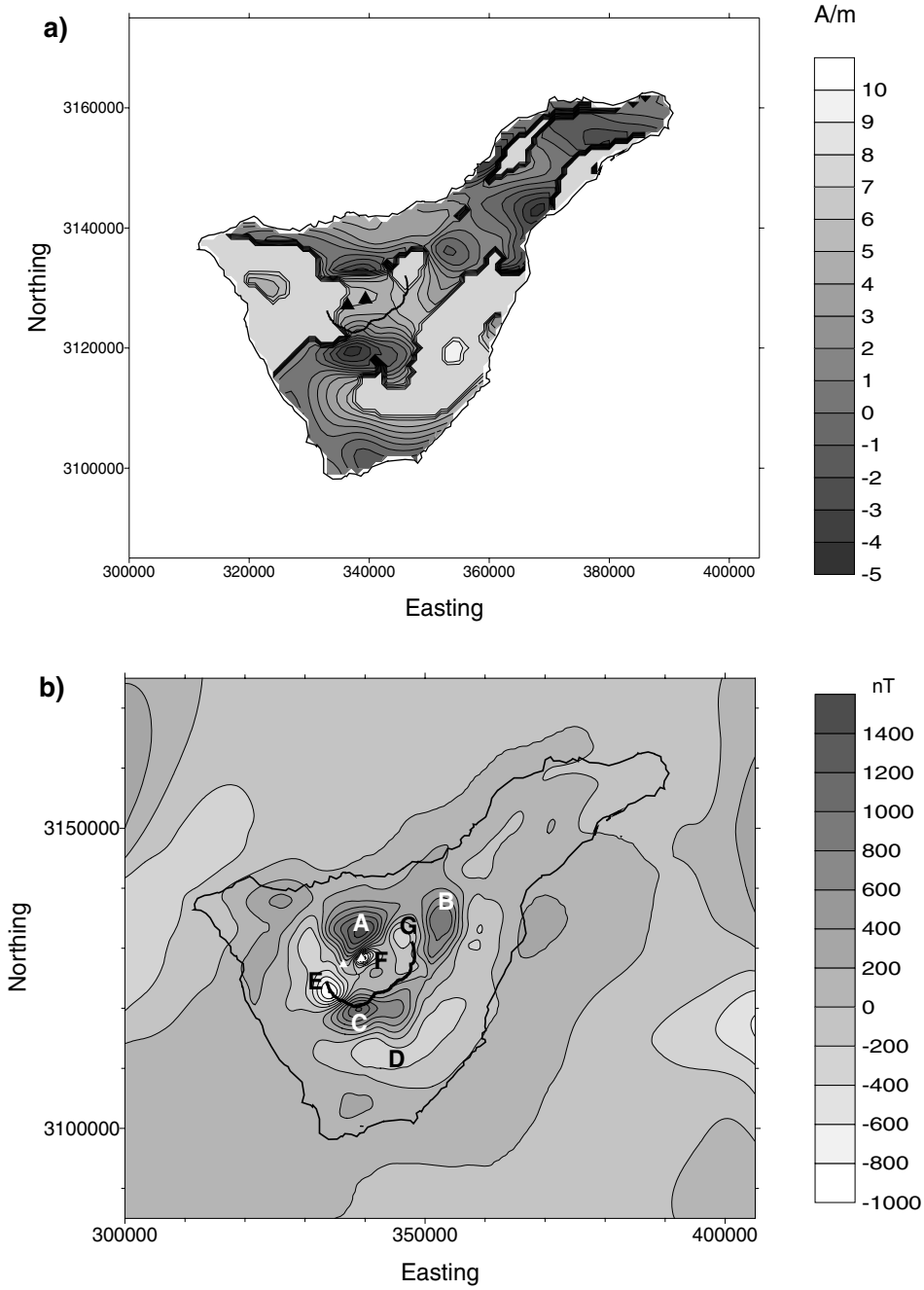

Fig. 9. (a) Distribution of shallow magnetisation obtained from the magnetic anomaly map by assuming minimum correlation between sources of interest and topographic features. The zero-magnetisation isoline is slightly thicker than the other ones to easily identify zones of negative magnetisation. The large clear areas correspond to places where the initial magnetisation considered $M_0 = 7.5$ A m^{-1} is valid. (b) Reduced-to-the-pole terrain-corrected magnetic anomaly map, obtained by removing from the original magnetic map (Fig. 4) the anomaly of the terrain characterised by the distribution of magnetisation of Fig. 9a. The most intense anomalies are identified.

(or regional) anomalies. Secondly, the magnetic anomaly map was low-pass filtered with a cut-off wave number $K_2 = 0.17\ km^{-1}$ ($\lambda = 5.8\ km$), since the spectrum shows that for higher wave numbers the magnetic sources are mainly subaerial. Therefore, the filtered map should contain the magnetic anomalies created by bodies located below sea level. The resulting filtered maps are shown in Fig. 8 where we distinguish anomalies produced by bodies located more than 5 km b.s.l. (Fig. 8a) and those produced by both deep and intermediate bodies located below sea level (Fig. 8b).

4.2.3. Terrain correction

After studying the different approaches developed for the resolution of the terrain correction problem (Blanco, 1997), the most promising method was considered to be the one developed by Grauch (1986, 1987). To apply the variable-magnetisation terrain correction method, we use a digital elevation model of the island, with a resolution of 1000 m. Then, the magnetisation grid was calculated using $M_0 = 7.5\ \mathrm{A\ m^{-1}}$ as initial magnetisation. This is a representative average value obtained through a statistical analysis of the magnetic anomaly map and it also coincides with the value used for these analysis in Hawaii (Hildenbrand et al., 1993). M_0 was determined by calculating the standard deviation of the synthetic topographic anomaly assuming $M = 1\ \mathrm{A\ m^{-1}}$ and dividing by the standard deviation of the original magnetic anomaly map, as proposed by Blakely and Grauch (1983). A proof of the reliability of the terrain correction method chosen is that the resulting surface magnetisation distribution (Fig. 9a) is consistent with the map of magnetic polarities of Tenerife published by Carracedo (1979).

The synthetic effect of the topography characterised by the obtained magnetisation distribution was calculated using Parker's algorithm (Parker, 1972) as implemented by Blakely (1981). The terrain magnetisation was assumed to be parallel to the present geomagnetic field, that is, $D = -8°$ and $I = 38°$. The terrain-corrected residual magnetic anomaly map was obtained by subtracting the synthetic terrain effect from the original magnetic anomaly map. To simplify interpretation, this map was reduced to the pole and the most interesting anomalies were identified (Fig. 9b).

5. Modelling

5.1. Gravity inversion model

The inverse gravity problem, namely the determination of a subsurface mass density distribution corresponding to an observed gravity anomaly, presents an intrinsic non-uniqueness for the solution (e.g. Al-Chalabi, 1971). Nevertheless, particular solutions can be obtained by including additional information about the model parameters (subsurface structure) and about the data parameters (statistical properties of the inexact data). Several methods have been developed to introduce a priori model hypothesis obtaining density contrast models, which fit the observed anomalies. In this paper, to investigate the distribution of inner anomalous masses and to establish possible relationship between anomalous density zones and volcanic process, we have investigated a three-dimensional (3D) distribution of density contrasts using gravity data and an inversion method developed and applied to other volcanic islands (Camacho et al., 1999).

For the present gravity study we have adopted a 3D partition of the local subsoil into 16,000 rectangular prisms, with sides ranging from 500 m (for the shallow elements) to 1500 m (for the deeper blocks). We selected a few possible density contrasts: −300, 0 and 200 kg m^{-3} for the whole subsoil volume. These values, smaller than those which can be deduced from a seismic model of Watts et al. (1997), have been selected empirically to obtain coherent volumes of the anomalous bodies. The obtained model depends on the adopted density contrast (for higher contrast, smaller volumes would be obtained), and we must keep in mind that the gravity inversion is nearly insensitive to horizontal mass stratification.

The solution for the gravity inversion is a 3D distribution of the adopted anomalous density contrasts plus a regional trend simultaneously adjusted. Figs. 10 and 11 show the obtained 3D model by means of several horizontal sections and vertical profiles. The adjusted regional component is a simple SE–NW trend. Gravity anomalies obtained by this solution reproduce the observed field, with residuals, obtained by means of the difference with observed anomalies, that follow a normal distribution with mean 0 mGal and standard deviation of 1.52 mGal that is coherent with the noise level that appears in the data (Fig. 6).

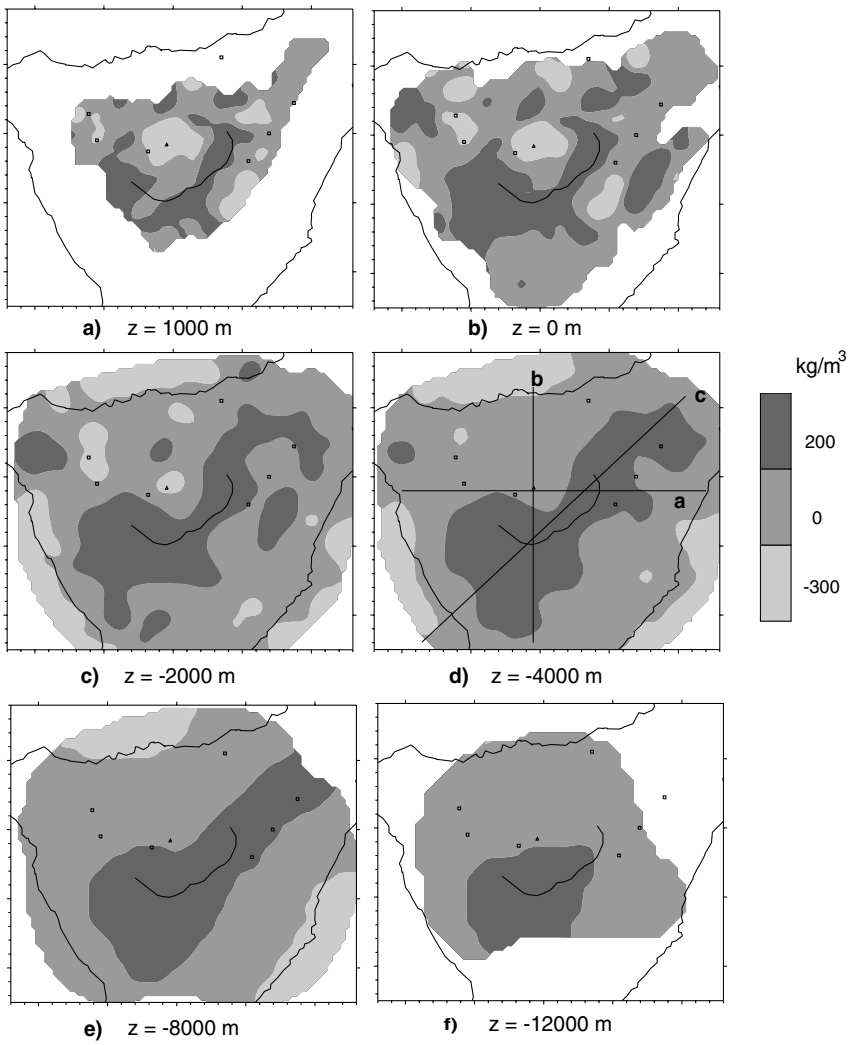

Fig. 10. (a to f) Horizontal sections of the adjusted 3D anomalous density model for several depths (z).

5.2. Structural magnetic models

A 2.5D forward modelling along two profiles crossing the main anomalies displayed in Fig. 8 was performed. The applied modelling technique calculates magnetic anomalies along a profile assuming that the source bodies are finite prisms of polygonal cross-section and uniform magnetisation, taking the expressions for these anomalies from Shuey and Pasquale (1973). In order to simplify the modelling process, all the selected profiles were produced by slicing the reduced-to-the-pole magnetic anomaly grids. Therefore, both the magnetic field and the magnetisation vectors are vertical.

First, the deepest structures whose effect is contained in the magnetic map were modelled. For this purpose, the low-pass filtered map with a cut-off wave number $K_1 = 0.05\ \mathrm{km}^{-1}$ (Fig. 8a) was used. As explained before, the spectral analysis of the magnetic anomaly map (Fig. 7) shows that this wave number marks a change in the slope of logarithm of the radially averaged power spectrum, so that the wave numbers $K < K_1$ characterise the deepest and regional sources, located below 5.7 ± 0.8 km. The

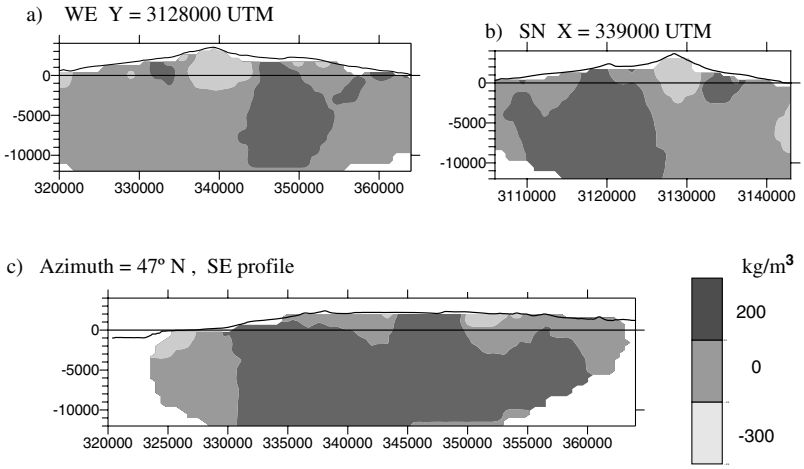

Fig. 11. Vertical profiles of the adjusted 3D anomalous density model across the sections pointed in Fig. 10d: (a) West–East, (b) South–North, close to the magnetics profiles of Fig. 8; and oblique profile for azimuth 47°, close to the ridge axis,

magnetic map that results from this filtering shows a large positive anomaly, elongated E–W (Fig. 8a). To model this anomaly, the filtered map was sliced along two perpendicular directions, one E–W and the other N–S, both intersecting the maximum of the anomaly. The model was constructed on the assumption that the anomaly was created by one body extending from 5–6 to 13–14 km below sea level (Fig. 12).

The low-pass filtered map for a cut-off wave number $K_2 = 0.17\ \mathrm{km}^{-1}$ (Fig. 8b) displays two intense positive anomalies superimposed on the broad high of deep origin enhanced in Fig. 8a. Therefore, starting from a model of the deep body previously obtained, the sources of these two magnetic highs were also modelled along the same profiles WE and SN. The resulting bodies extend from the top of the deep structure at about 5–6 km b.s.l. to 1.5–2.0 km a.s.l. (Fig. 13).

Finally, the terrain-corrected residual magnetic map was used to analyse the shallowest structures. In this map, on the contrary to the previous ones, several negative anomalies (E, F, and G in Fig. 9b) can be appreciated.

6. Interpretation and discussion of the anomalies

In order to interpret the observed gravity and magnetic anomalies, we have defined three zones at different depths in agreement with the established models. The limits of these zones coincide approximately with some inversion sections of the gravity model (Fig. 10) and with the magnetic horizons defined according to the spectral analysis (Fig. 7) of the magnetic anomaly map: (i) deep zone (below 12 km); (ii) intermediate zone (between 12 km b.s.l. and 1 km a.s.l.); and (iii) surface zone (from 1 km a.s.l.). The deep zone corresponds to the basement of the island. Most of the intermediate zone corresponds to the submarine shield building stages. Most of the surface zone corresponds to the big central edifice Cañadas–Teide. A magnetic horizon, at 5.7 ± 0.8 km b.s.l., permits the separation of two levels in the intermediate zone.

6.1. Deep zone

6.1.1. Gravity

In Fig. 10f an anomalous structure is clearly distinguished whose steep slope defines a large source with a well defined geometry at 12 km depth. We suggest that this anomaly is due to an uplifted block of the basement in the same way as other gravity anomalies under large volcanoes have been interpreted and corroborated by different geophysical techniques (Villaseñor et al., 1998; Barberi et al., 1994). The elevation of the basement blocks can be inferred from the deformation of the geoid under Tenerife (Sevilla et al., 1999) and is clearly shown beneath

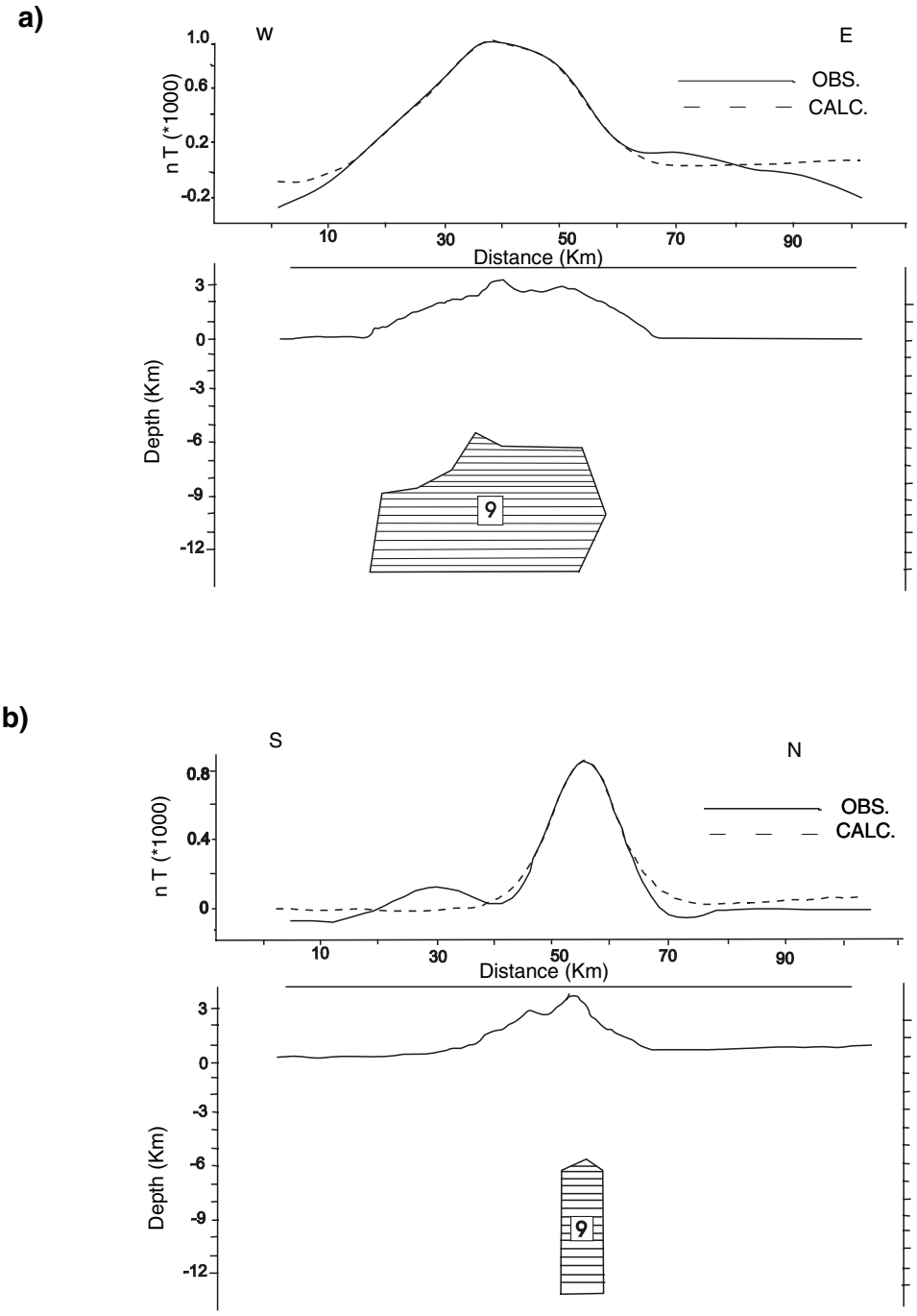

Fig. 12. Forward modelling of the long-wavelength magnetic anomaly of Fig. 8a. The number written on the source is the intensity of magnetisation in A m^{-1}. Both the magnetic field and the magnetisation are supposed to be vertical, as the anomalies have been reduced to the pole. The black line is the observed anomaly while the dashed line corresponds to the synthetic anomaly created by the model. The elevation profile is also shown: (a) model along WE profile; (b) model along SN profile.

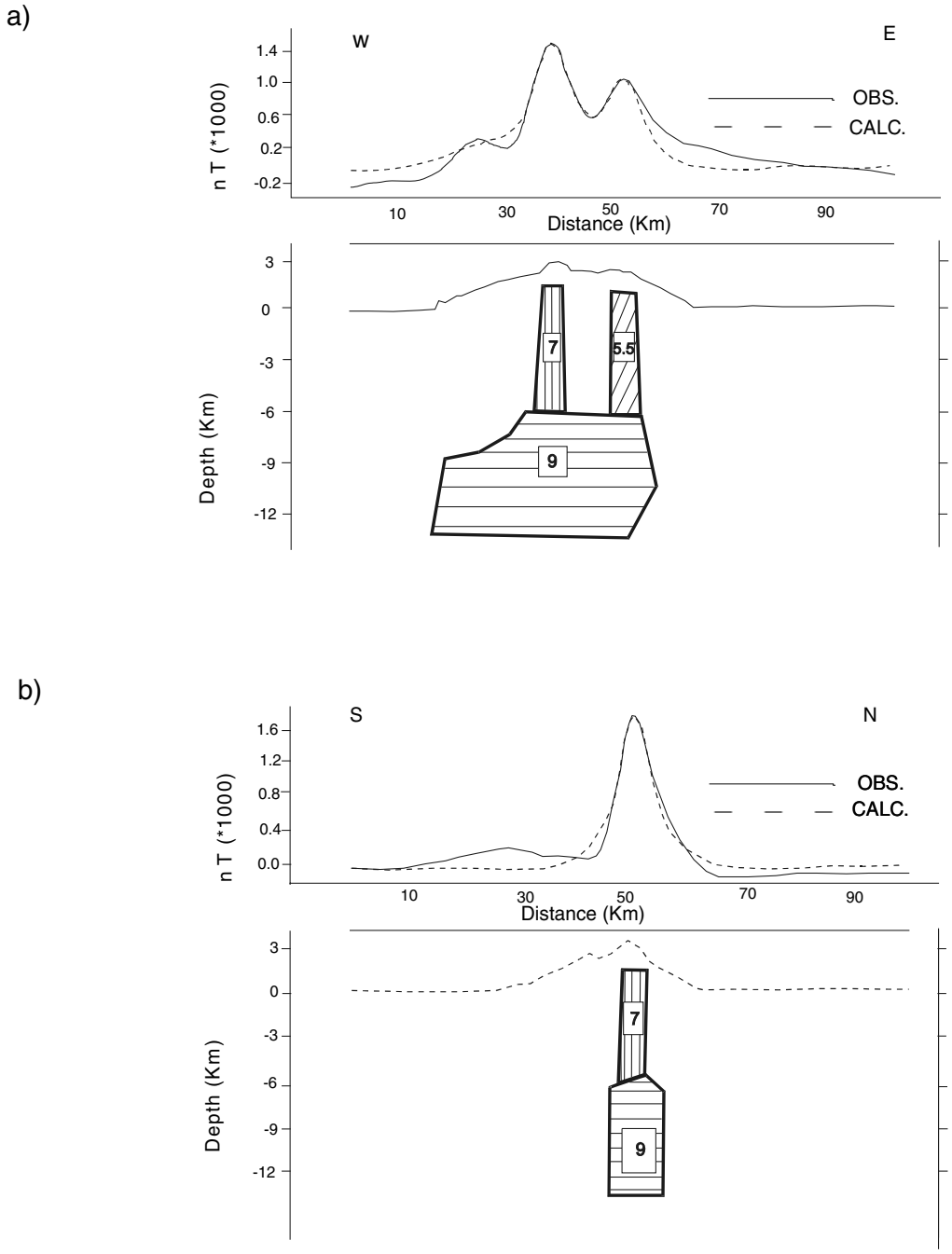

Fig. 13. Forward modelling of the magnetic anomalies created by the structures located below the sea level of Fig. 8b. The number written on the sources is the intensity of magnetisation in A m^{-1}. The black line is the observed anomaly while the dashed line corresponds to the synthetic anomaly created by the model. The elevation profile is also shown. The models are along WE and SN profiles in Fig. 8b that coincides with the same profiles in Fig. 8a.

other islands of the Canary archipelago where large tilted blocks have been differentially uplifted from the sea floor (Araña and Ortiz, 1991; Marinoni and Pasquaré, 1994). The amount of uplift ranges from 2 km in La Palma (Staudigel and Schmincke, 1984) to more than 4 km in Fuerteventura (Robertson and Stillman, 1979). Araña and Ortiz (1991) attribute this uplift to the action of faults with important dip slips, while Fernández et al. (1997) suggest a shear tectonic process.

Alternatively, this anomaly could be produced by the existence of underplated material at the lower crust although this possibility has been rejected by Watts et al. (1997). Further, this anomaly could be interpreted as the result of intrusion of dense mafic magma, now transformed in dense gabroic bodies, but the geometry and volume of the source seems to be too big to fit this hypothesis. Neither does it seem to be realistic to model a giant shallow body that requires a density as high as 3210 kg m^{-3} (Ablay and Kearey, 2000).

6.1.2. Magnetism

The magnetic anomaly map of Tenerife contains anomalies ranging in wavelength from 2 to about 57 km. Spectral analysis of the anomalies indicates that the regional anomalies are those characterised by wave numbers lower than 0.05 km^{-1}, which is equivalent to a wavelength of 20 km. The sources of these anomalies are structures whose bottom is located at depth less than 16 km, as deduced from the dimensions of the map (Shuey et al., 1977). This figure fits with the location of the Moho between 13 and 15 km, after different geophysical works (Bosshard and MacFarlane, 1970; Banda et al., 1981; Ortiz et al., 1986; Suriñach, 1986; Watts et al., 1997).

6.2. Intermediate zone

6.2.1. Gravity

At 8 km depth (Fig. 10e), a strong anomalous structure with a NE–SW trend is shown. The bottom of this body could be deeper if we adopt a lesser density for the sediments lying on top of the oceanic crust. However, in accordance with our interpretation of the deep zone, the source of this anomaly overlies oceanic layer 3, whose top has also been located at about 8 km b.s.l. by Watts et al. (1997).

The SW end of the anomaly is superposed with the anomaly that we have associated with the deep zone. Nevertheless, a greater extension towards the South coincides on the surface of the island with the remains of the Roque del Conde massif (Fig. 1). We suggest that the source of this anomaly is the base of an elongated basaltic edifice corresponding to the initial stage of the Tenerife shield. This construct extends from Roque del Conde towards the NW with a NE–SW axis. This direction coincides with that of a fracture system that has been active in the whole Archipelago in different constructive periods, according the lithosphere stress regime (see, e.g. Mezcua et al., 1992; Marinoni and Pasquaré, 1994). Watts (1994) calculates a flexural load of about 2 km underneath the central region of Tenerife, while Funk et al. (1996) situate the base of the Anaga shield at about 5 km b.s.l. These figures fit with the beginning of the submarine volcanism of Roque del Conde over a sea floor that is today found at about 8 km depth. The shield development is better appreciated at 4000 m depth (Figs. 10d and 11) where the base of the Teno edifice is also seen. Probably, Anaga (not seen in Figs. 10 and 11) arose as another independent edifice in the NE prolongation of the SW–NE axis at about 5000 m depth (Funk et al., 1996).

The old massifs are indicated in the Bouguer anomalies map (Fig. 4) with an intensity proportional to their dimensions and superficial extension. However, the big elongated edifice that constituted the first stage and core of the submarine shield (Figs. 10b–d and 11a and c) halted its growth, or was destroyed at its NE edge, being practically buried by post-shield volcanism.

The minima that appear from 2000 m b.s.l. become more intense at shallow depths and mainly in the superficial zone, indicating its association with recent volcanic episodes. The minimum off-shore, parallel to the Northern coast (Fig. 10c and d) is probably an edge effect.

6.2.2. Magnetism

The most evident feature of the reduced-to-the-pole long-wavelength magnetic map (Fig. 8a) is an intense positive anomaly localised at the northern edge of the gravity high that we have associated with an uplifted

block of the basement. We cannot relate this magnetic anomaly with the shield materials that should have a very low degree of magnetisation due to their age and submarine environment (Prévot and Grommé, 1975). Forward modelling of this anomaly (with an E–W orientation) resulted in a prismatic body situated between 5–6 and 12–14 km depth, characterised by a strong magnetisation contrast of 14 A m^{-1} and about 40 × 8 km in the E–W and N–S directions, respectively (Fig. 12). We suggest that this intense magnetisation is due to the presence of gabbro-ultramafic cumulates associated with the dyke swarm that fed the core of the submarine shield and later the magma chambers of the central edifice. This kind of cumulates (rich in magnetite, ilmenite, pyrite and other magnetic matter) are typical of the Basal Complexes of Gomera and Fuerteventura (see, e.g. Stillman, 1987) and crop out locally at the base of the Anaga massif (Rodríguez-Losada and Martínez-Frías, 1998). However, these accumulates and the filonian network would do not constitute a massive body because they are very fractured and therefore could not produce any gravity high.

Two intense positive anomalies superimposed over the linear long-wavelength anomaly of Fig. 8a. The source of the anomaly below Teide–Pico Viejo (labelled T in Fig. 8b) was quantified in terms of geometry and magnetisation contrast along the same profiles used to identify the deep source (Fig. 13). The most appropriate and simple geometry happened to be a vertical prism characterised by a magnetisation of 7 A m^{-1}. This prism can be understood as the feeding conduits of basaltic eruptions that supplied the construction of the Teide–Pico Viejo volcanoes. The existence of these dike systems has also been proposed by Banda et al. (1981) and Watts et al. (1997) to explain an unusually high seismic velocity zone beneath central Tenerife. Moreover, the existence of shallow magma intrusions has also been evidenced in a study of small earthquakes recorded on Las Cañadas caldera (Del Pezzo et al., 1997).

The northern part of the anomaly P in Fig. 8b coincides with the Pedro Gil volcano that is the only one on the island big enough to be compared with Teide or Pico Viejo. So, a similar prismatic body could be used to model the source below Pedro Gil as the profile WE in Fig. 13 suggests. The weak magnetisation difference (7–5.5 A m^{-1}) between the two prisms could be explained in terms of age (<170 ka Teide and >800 ka Pedro Gil). Further, the Pedro Gil volcano presents an important dike network at the surface.

In Fig. 8b, three other positive anomalies, much less intense than T and P, can be observed. They are displayed to the NW (Teno massif), to the NE (Anaga massif) and to the S of the island (Roque del Conde), where the strong weathering could explain the weak magnetisation that causes the smooth positive anomalies displayed over these areas.

6.3. Superficial zone

6.3.1. Gravity

For shallow levels, the massive volcanic core adopts a more complicated morphology. The shallowest sections for the area of the Las Cañadas caldera show a density minimum surrounded by a rhomboidal high density structure, with a very light north component or break (Fig. 10a and b). In Fig. 10a we also observe that in the Ucanca zone (see Fig. 1), an elongated gravity minimum coincides with a part of an alignment (139° N) that breaks the SW caldera margin. The alignments detected in this central area agree with the directions suggested by Ablay and Martí (1995) as main eruptive alignments.

We agree with Vieira et al. (1986) and Camacho et al. (1991), that in former microgravimetric studies of the Las Cañadas caldera, interpreted part of this feature as the infilling sequence of a collapsed caldera, following the model of Yokoyama (1974). Such a model could be at present upheld by the western sector of the caldera (Ucanca), but the NE sector would have reduced dimensions between the present wall (Diego Hernández) and the eastern flank of the Teide. Alternatively, the high-density materials identified within the present caldera boundaries can be related to an intracaldera extension of the ridges, as suggested by Ablay and Kearey (2000). The group of density lows (Fig. 10a and b) mostly correlate with recent volcanism. The central minimum is confined to the Teide–Pico Viejo–Montaña Blanca complex and represents the accumulation of volcanic light materials that constitute the cone. These light materials reached a maximum depth of 2 km (Fig. 11a and b) for the adopted negative density contrast (−300 kg m^{-3}).

6.3.2. Magnetism

The interpretation of the magnetic anomalies linked to the shallow structures which conform the recent (<3 Ma) volcanic edifices of Tenerife requires to remove the magnetic effect resulting the terrain-corrected reduced-to-the-pole residual map (Fig. 9b) that evidences several short-wavelength anomalies that mark the boundaries of the central edifice. The most interesting anomalies are those negative ones, since they have no relation with the deep and intermediate structures described before. The upper magnetic horizon detected at about 1400 m a.s.l. (Fig. 7a) can be explained as the transition between the basaltic materials, which make up an old massif in the central part of the island, and the phonolitic rocks of the Las Cañadas edifice that buried them. This unconformity together with those ones affecting the overlaying materials can be the sources of the shallow magnetic anomalies, taking into account that a greater depth of burial results in a larger attenuation of short wavelength anomalies.

Positive anomalies A and B in Fig. 9b are very similar to the positive anomalies (T and P) in Fig. 8b and we assume that they are caused by the same sources; that is, the basaltic dike networks mentioned above (intermediate upper zone), which in some occasions reach the surface, as occurs in the area of the Pedro Gil edifice. Also, the magnetic effect of the basaltic volcanics erupted by the Teide should contribute to anomaly A, as a big mass of basalts flowed to the North producing an accumulation of inflated sheet flows filling a broad valley where large lava tunnels formed. Hildenbrand et al. (1993) speculated that such accumulations of inflated sheet flows should be related to intense magnetic highs. We have not enough data to speculate about the positive anomaly C that appears over the southern boundary of the Las Cañadas caldera. A possible source of this magnetic high could be a normally polarized terrain that predates or intrudes on the Cañadas Series.

In the central part of the island three mangetic lows appear (Fig. 9b). One of them (E) occupies the western boundary of the Las Cañadas caldera, while the two others (F, G) are located inside the caldera. Anomaly F appears just over the Teide cone, while anomaly G practically coincides with the last caldera collapse (<0.2 Ma, according to Martí et al., 1994). In the three cases, its short wavelength and high amplitude indicate shallow sources. These anomalies may be considered to form a continuous low, locally interrupted, in the present caldera of the Las Cañadas extending along NW–SE volcanic ridge. In the discussion of the origin of this non-magnetised zone, there are several different possible explanations to justify the different intensity and shape of each individual anomaly. To explain the pronounced low of anomaly F, we could consider the presence of a magma reservoir, but it is improbable to find magma so close to the surface. Another possible origin would be the presence of intensely hydrothermally altered materials, as this type of alteration usually oxidates the ferromagnetic minerals, drastically reducing the remanent magnetisation of the rocks. In fact, the Teide summital fumaroles show the interaction between the magmatic gases rising up from a shallow magma chamber and the local aquifer that exists within the Teide cone at about 2500–3000 m a.s.l. (Albert-Beltrán et al., 1990). Therefore, it is likely that the volcanics within the Teide cone are affected by some hydrothermal alteration linked to the presence of fluids at high temperature. This seems to be a reliable explanation for a null magnetisation inside the Teide cone and coincides with the gravity model before discussed. In fact, the destruction of magnetite by hydrothermal alteration and its effect on the magnetisation of the rocks has been observed in several volcanic areas and considered to be the cause of magnetic lows (e.g. Kane et al., 1976; Rapolla et al., 1989; Hildenbrand et al., 1993; Blanco et al., 1997).

Anomaly G is different, as there is no evidence of high temperatures close to the surface in this zone. An accumulation of phonolitic materials infilling a buried valley or a collapse caldera would explain the origin of this negative anomaly, as this kind of lithology is always associated with very low magnetisation compared with basalts. The caldera wall (Diego Hernández) to the south of this anomaly is mainly formed by pumice deposits, which are probably thicker inside the caldera basin as suggested by a micro-seismic survey carried out in this zone (Del Pezzo et al., 1997). Anomaly E could be interpreted as being created by a reversed polarity of a negative structure located beneath the western caldera rim since its amplitude and size require a very strong

magnetisation contrast. Alternatively, we could speculate that these anomalies (G-E) constitute a continuous line, slightly displaced and widened with respect to its sources, that could be associated with a set of collapse calderas having different features in its eastern and western sectors.

Finally, anomaly D, located outside the caldera, over the south slopes of the Cañadas Edifice, seems to be caused by a complex source when the shape of the isolines is taken into account together with a comparable minimum gravity trend. In this case, we have not any specific explanation except litologic contrast in buried slopes of the central and ridge edifices.

7. Conclusions

Our interpreted study of gravity and magnetic anomalies of Tenerife has improved the knowledge about the internal structure beneath this island. A high density and extensive body, probably corresponding to an uplifted block of oceanic crust has been identified. The initial (Roque del Conde bottom) and the late (Anaga bottom) stages of the submarine shield of Tenerife have grown under a NE–SW tectonic control. The central sector of this submarine shield is buried under the central region of the island. The NE–SW tectovolcanic axis remained active, but other (mainly NNW–SSE) trends appeared in successive stages of the shield development, constructing Teno and probably other buried edifices. The base of the shield core is presently located at about 8-km b.s.l. The old basaltic massifs of Teno (NW), Anaga (NE) and Roque del Conde (S) are reflected in the Bouguer anomalies map as gravity highs and in the magnetic anomaly map as smooth magnetic highs. The magnetic properties of the b.s.l. materials (submarine volcanics and sediments very demagnetised over millions of years) present a clear contrast with the overlaying beds.

A highly magnetic intrusive body characterised by an E–W linear geometry has been detected beneath central Tenerife at a depth between 5–6 and 13–14 km. This body could be produced by gabbro-ultramafic cumulates associated with the voluminous and repetitive magma feeding systems that construct the core of the shield stage and the central edifice of Tenerife. The magnetic anomaly map also reveals two vertical dike complexes strongly magnetised with normal polarity, which extend from the top of the high-magnetic body. These conduits have supplied the construction of the recent basaltic edifices Teide–Pico Viejo and Pedro Gil, over which strong magnetic anomalies are displayed. Weakly magnetised rocks belonging to the basaltic shield stages flank both deep and intermediate intrusions. In the central part of the island, the gravity and magnetic anomalies produced by above sea level sources could indicate the existence of a collapse caldera system filled by low density and low magnetic products. The spectral analysis of the magnetic anomaly map revealed two magnetic interfaces that were interpreted as the top of the deep intrusive bodies or magma chambers zone (≈ 5.7 km b.s.l.) and the bottom phonolites of Las Cañadas edifice (≈ 1.4 km a.s.l.). The subaerial part of Tenerife presents a magnetic and gravimetric complexity due to the contrast between the high density and magnetic intensity of recent basalts and the low density and magnetic intensity of the Central edifice phonolites whose first levels also present a reversal polarity. Hydrotermal alteration is assumed to be the cause of a short wavelength magnetic low over the Teide volcano.

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